CSE400 M2 GROUP- 8

NETWORKING

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Project Context and Motivation

Title: Task Offloading in D2D Edge Networks

Domain: Wireless Networking & Edge Computing

Context: High-density mobile networks (5G/6G)

Problem: Mobile CPUs are overwhelmed by latency-sensitive traffic

Intrinsic Uncertainty: Wireless channel fading and random packet arrivals

Stochastic Need: Deterministic routing fails under network jitter

Project Objective and Scope

Objective: Maximize Global Success Expectation: $\max_a \sum_{i \in N} E[Y_i]$

Decision: Optimal offloading target selection in a mesh

Prediction: Outage probability (Probability of missing deadline τ)

Scope: Distributed D2D network with shared compute-resources

Metric: Moving from "Average Delay" to "Bernoulli Success Probability" (Yi ~ Bernoulli(pi)).

Modeling Goal: Maximize the expected number of tasks meeting strict deadlines τ_i.

Project System Overview

Pipeline: Packet Arrival → Offloading Choice → Wireless Link → Remote Buffer

Major Components: Initiator Nodes, Service Peers, and Wireless Channels

Data Flow: Task metadata bits transmitted over a stochastic medium

Interaction: Stochastic Interaction: Aggregate Arrival Rate $\lambda_{\square} = \sum_{\square} a_{\square,\square} \lambda_{\square}$ (Neighbor Load).

System Flow: Offloading Decision → Discrete Summation → M/M/1 Sojourn Time → Binary Outcome.

Sources of Uncertainty

Channel Uncertainty: Small-scale fading and shadowing in wireless links

Traffic Uncertainty: Burstiness of packet arrival times (Poisson jitter)

Processing Uncertainty: Variable CPU cycles required for network protocols

Load Uncertainty: Hidden states of peer device queues

Key Random Variables

Xproc : Task Processing Time (Continuous Random Variable).

Tinter : Inter-arrival Time ~ Exponential(λ).

Wq : Buffer Delay (Derived Random Variable).

Lq : Queue Length ~ Geometric Distribution (Networking Metric).

B / H : Bandwidth Demand / Channel Link Quality.

Tsojourn : Total Delay (Aggregate Random Variable: Wq + Xproc + ttrans).

Yi : Completion Indicator (Success = 1, Fail = 0).

Probabilistic Models and Assumptions

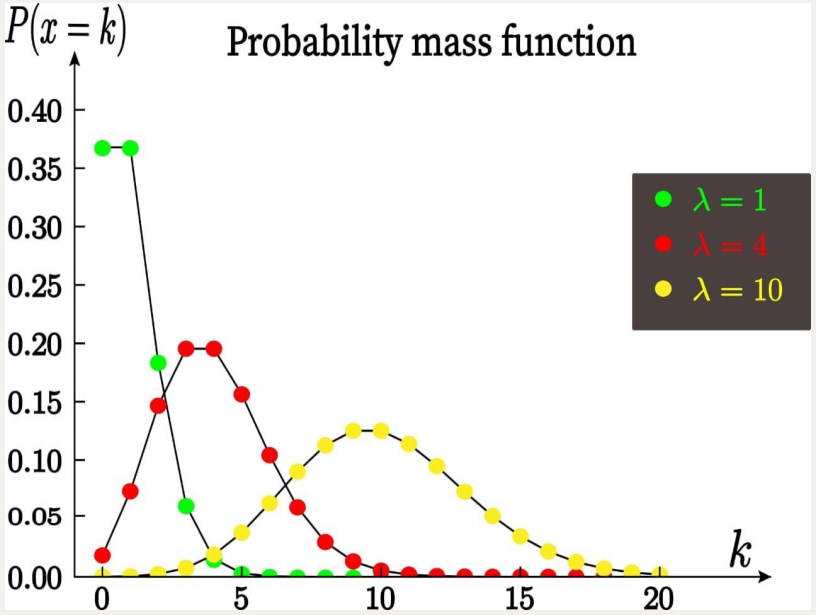
Arrival Model: Poisson Process with rate λ

Service Model: Exponential Service times with rate μ

Fading Assumption: Rayleigh Fading (Exponential power gains) influencing ttrans.

Independence: i.i.d. (Independent and Identically Distributed) links

Justification: Urban D2D environments follow stochastic multipath patterns



Probabilistic Reasoning and Inference Logic

Logic: Link quality γ determines the stochastic transmission rate R

Conditional Prob: P(Success | Neighbor Load)

Inference: Using Load λ to calculate the CDF of Sojourn time

Formula: $F(t \leq \tau) = 1 - \exp(-(\mu - \lambda) \cdot \tau_{eff})$

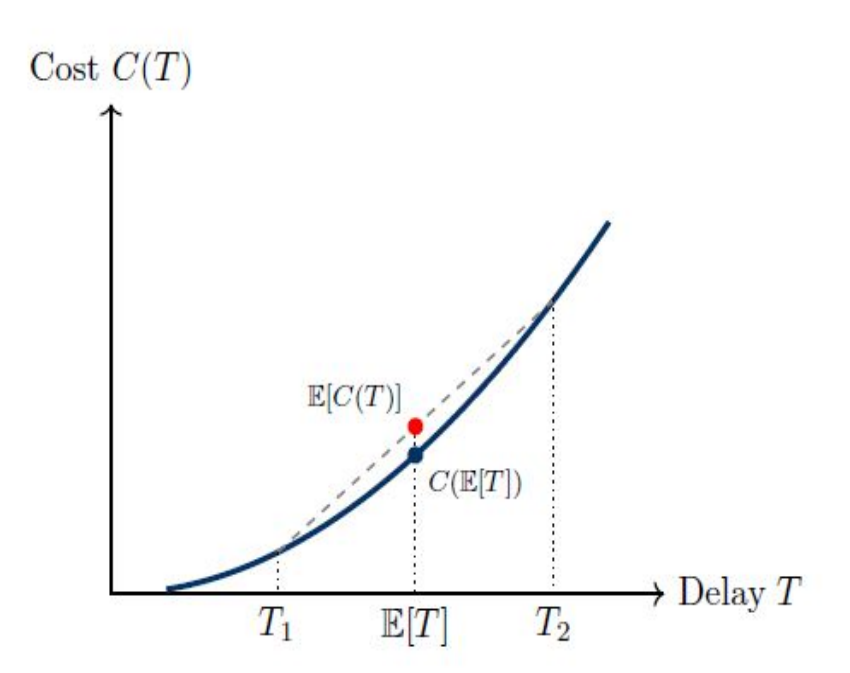
Operationalizing Probabilistic Reasoning

Simulation: Monte Carlo realizations of random packet streams

Jensen's Inequality: E[C(T)] ≥ C(E[T]).

Decision Logic: Offload if P(Success remote) > P(Success local).

Metrics: Percentage of packets meeting the Hard Deadline τ



Current Limitations and Conceptual Gaps

Mobility Gap: Model assumes static nodes; ignores Doppler fading

Feedback Delay: Uncertainty in "Stale" channel state information

Ambiguity: Handling highly correlated traffic bursts (Non-Poisson)

Feasibility: High computational overhead of calculating Nash Equilibrium

Planned Refinements and Role Coordination

Evolution: Upgrading $M / M / 1 \rightarrow M / G / 1$ for heavy-tailed risks

Refinement: Upgrading to M/G/1 queueing using Laplace–Stieltjes transforms for non-exponential heavy-tailed tasks.

Ambiguity Resolution: Solving the Stale Information problem

Project Manager (M3): Falak Shah

Scribe Team (M3): Ishani Prajapati, Nikunj Kalariya

Simulation/Video (M3): Manthan Pancholi, Urvil Chodvadiya

Test Goal: Injecting bursty traffic to stress-test Poisson assumptions

Summary of Group-Level Understanding

- Replaced deterministic fixed delays with stochastic random variables
- Established M/M/1 queueing model as the baseline system model
- Identified variance as a major risk factor, not just mean delay
- Modeled user offloading decisions as a stochastic game
- Showed system converges to a stable Nash equilibrium
- Used probability to quantitatively manage network congestion